3.1.2 Hot desert systems and landscapes

This section of our specification focuses on drylands which occur at all latitudes and are characterised by limited soil moisture caused by low precipitation and high evaporation. The focus is on hot deserts and their margins, where the operation of characteristic aeolian and episodic fluvial processes with their distinctive landscape outcomes are readily observable. In common with water and carbon cycles, a systems approach to study is specified.

Student engagement with subject content fosters informed appreciation of the beauty and diversity of deserts and the challenges they present as human habitats. The section offers the opportunity, in the right settings, to exercise and develop geographical skills, including observation, measurement and geospatial mapping skills, together with data manipulation and statistical skills, including those associated with and arising from fieldwork.

3.1.2.1 Deserts as natural systems

Systems in physical geography: systems concepts and their application to the development of desert landscapes – inputs, outputs, energy, stores/components, flows/transfers, positive/negative feedback, dynamic equilibrium. The concepts of landform and landscape and how related landforms combine to form characteristic landscapes.

The global distribution of mid and low latitude deserts and their margins (arid and semi-arid).

Characteristics of hot desert environments and their margins: climate, soils and vegetation (and their interaction). Water balance and aridity index.

The causes of aridity: atmospheric processes relating to pressure, winds, continentality, relief and cold ocean currents.

3.1.2.2 Systems and processes

Sources of energy in hot desert environments: insolation, winds, runoff.

Sediment sources, cells and budgets.

Geomorphological processes: weathering, mass movement, erosion, transportation and deposition.

Distinctively arid geomorphological processes: weathering (thermal fracture, exfoliation, chemical weathering, block and granular disintegration).

The role of wind – erosion: deflation and abrasion; transportation; suspension, saltation, surface creep, deposition.

Sources of water: exogenous, endoreic and ephemeral; the episodic role of water; sheet flooding, channel flash flooding.

3.1.2.3 Arid landscape development in contrasting settings

Origin and development of landforms of mid and low latitude deserts: aeolian – deflation hollows, desert pavements, ventifacts, yardangs, zeugen, barchans and seif dunes; water – wadis, bahadas, pediments, playas, inselbergs.

The relationship between process, time, landforms and landscapes in mid and low latitude desert settings: characteristic desert landscapes.

3.1.2.4 Desertification

The changing extent and distribution of hot deserts over the last 10,000 years. The causes of desertification – climate change and human impact; distribution of areas at risk; impact on ecosystems, landscapes and populations. Predicted climate change and its impacts; alternative possible futures for local populations.

3.1.2.5 Quantitative and qualitative skills

Students must engage with a range of quantitative and relevant qualitative skills, within the theme landscape systems. These should include observation skills, measurement and geospatial mapping skills and data manipulation and statistical skills applied to field measurements.

3.1.2.6 Case studies

Case study of a hot desert environment setting to illustrate and analyse key themes set out above and engage with field data (exemplifying field data may be gathered in settings that experience some of the aeolian processes associated with mid and low latitude desert environments such as coastal dunes).

Case study at a local scale of a landscape where desertification has occurred to illustrate and analyse key themes of desertification, causes and impacts, implications for sustainable development. Evaluation of human responses of resilience, mitigation and adaptation.